

Q.1 Which of the following compounds will show highest acidic strength?

1. 2,4,6-Trinitrophenol
2. Phenol
3. 2,4-Dinitrophenol
4. 2-Nitrophenol

Options 1. 1

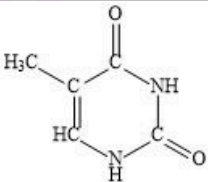
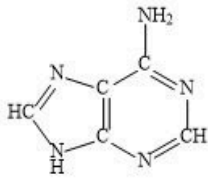
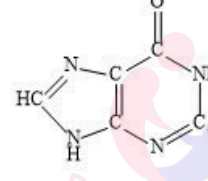
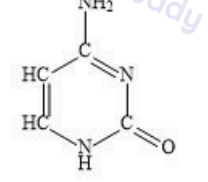
2. 2

3. 3

4. 4

Q.2

Match **List-I** (Name of the Base) with **List-II** (Structure)

List-I	List-II
Name of the Base	Structure
(A). Adenine (A)	(I). 
(B). Cytosine (C)	(II). 
(C). Thymine (T)	(III). 
(D). Guanine (G)	(IV). 

Choose the **correct** answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 The half life of a radioactive element is 140 days. After 560 days, one gram of the element will be reduced to:

1. 0.5 g
2. 0.0625 g
3. 0.125 g

4. 0.25 g

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Which transition metal ion is most stable against oxidation to its +3 state?

1. Mn^{2+}
2. Fe^{2+}
3. Cr^{2+}
4. Fe

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The cell emf depends on:

1. Size of anode
2. Size of cathode
3. Temperature
4. Volume of electrolytic solution

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Match List-I with List-II

List-I	List-II
Order of reaction	Unit
(A). Zero order reaction	(I). $\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$
(B). First order reaction	(II). $\text{mol}^{-1} \text{L s}^{-1}$
(C). Second order reaction	(III). $\text{mol L}^{-1} \text{s}^{-1}$
(D). Third order reaction	(IV). s^{-1}

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which of the following statement(s) is correct ?

- (A). Keratin is an example of fibrous protein.
- (B). Fibrous proteins are water-soluble, while globular proteins are water-insoluble.
- (C). Polypeptide chains in fibrous proteins run parallel and are held together by hydrogen and disulphide bonds.
- (D). Insulin and albumins are common examples of globular proteins.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C), and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The rate of a chemical reaction doubled for every 10°C rise in temperature. If the temperature is raised by 50°C , the rate of reaction will be

1. 50 times higher
2. 32 times higher
3. 40 times higher
4. 20 times higher

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Match List-I with List-II

List-I List-II

Laws/properties Statements

- | | |
|---------------------------------------|--|
| (A) Colligative properties | (I) for a solution of volatile liquids, the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in the solution. |
| (B) Henry's law | (II) the total pressure over the solution phase in the container will be the sum of the partial pressures of all the components of the solution. |
| (C) Raoult's law | (III) depends on the number of solute particles, irrespective of their nature. |
| (D) Dalton's law of partial pressures | (IV) at a constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of a liquid or solution. |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (I), (B) - (III), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

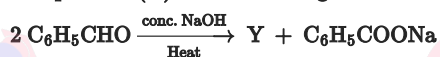
Q.10 Which of the following pairs of solutions are isotonic?

1. 0.1 M NH_2CONH_2 and 0.1 M NaCl
2. 0.1 M NH_2CONH_2 and 0.1 M MgCl_2
3. 0.1 M $\text{Ca}(\text{NO}_3)_2$ and 0.1 M Na_2SO_4
4. 0.1 M NaCl and 0.1 M $\text{C}_6\text{H}_{12}\text{O}_6$

Options 1. 1

2. 2
3. 3
4. 4

Q.11 The product (Y) in the following reaction is



1. $\text{C}_6\text{H}_5\text{COOH}$
2. $\text{C}_6\text{H}_5\text{CH}_3$
3. $\text{C}_6\text{H}_5\text{OH}$
4. $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$

Options 1. 1

2. 2
3. 3
4. 4

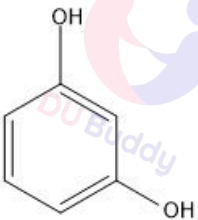
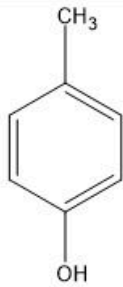
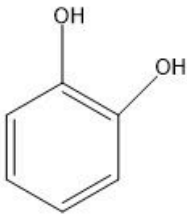
Q.12 Alkyl halides react with sodium to give alkanes with double number of carbon atoms in

1. Sandemeyer's reaction
2. Wurtz reaction
3. Wurtz-Fittig reaction
4. Williamson synthesis

Options 1. 1

2. 2
3. 3
4. 4

Match **List-I** with **List-II**

List-I	List-II
Name of compound	Structure
(A). Resorcinol	(I). 
(B). Catechol	(II). 
(C). Hydroquinone	(III). 
(D). p-Cresol	(IV). 

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (VI), (B) - (III), (C) - (II), (D) - (I)

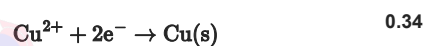
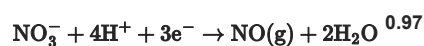
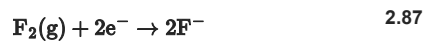
Options 1. 1

2. 2

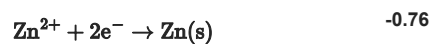
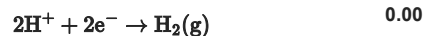
3. 3

4. 4

Q.14 Based on the standard electrode potential given in the table
Half-cell reaction (E°/V) at 25°C)



Half-cell reaction (E-/V) at 25°C)



Choose the wrong / incorrect statement from the options given below:

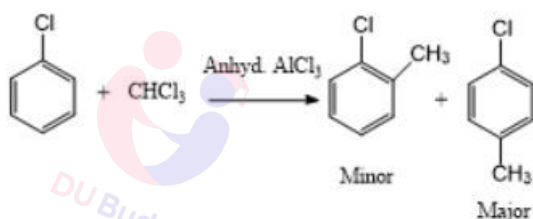
1. Fluorine gas is a strong oxidizing agent.
2. Lithium ion is a strong oxidizing agent.
3. Copper can be oxidized by nitrate ion.
4. Zinc can be oxidized by hydrogen ion.

Options 1. 1

2. 2
3. 3
4. 4

Q.15

Friedel-Craft reaction as shown below is an example of



1. Electrophilic substitution reaction
2. Nucleophilic substitution reaction
3. Electrophilic addition reaction
4. Nucleophilic addition reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.16 The molecule with largest C-X bond length is

1. $\text{CH}_3\text{-Cl}$
2. $\text{CH}_3\text{-I}$
3. $\text{CH}_3\text{-Br}$
4. $\text{CH}_3\text{-F}$

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Which of the metal is used in catalytic hydrogenation?

1. Cd
2. Ni
3. Fe
4. Co

Options 1. 1

2. 2
3. 3
4. 4

Q.18 What is the correct order of following steps in the preparation of potassium permanganate?

- (A) disproportionation in a neutral or acidic solution
- (B) production of dark green K_2MnO_4
- (C) MnO_2
- (D) fusion with alkali metal hydroxide and KNO_3

Choose the \texttt{correct} answer from the options given below:

1. (B), (A), (C), (D)
2. (C), (D), (B), (A)

3. (A), (B), (D), (C)

4. (D), (B), (C), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Catalytic hydrogenation of ketones with lithium aluminium hydride gives

1. Primary alcohols
2. Secondary alcohols
3. Tertiary alcohols
4. Esters

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Choose the correct statements from the following related to glycine amino acid.

(A). Water soluble α -amino acid.

(B). It has both amino and carboxylic groups in its structure.

(C). It is optically inactive.

(D). It is a non-essential amino acid.

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D)

2. (A), (B) and (C) only

3. (A), (C) and (D) only

4. (B), (C) and (D) only

Options 1. 1

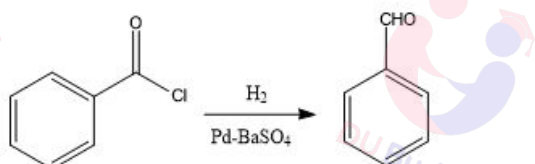
2. 2

3. 3

4. 4

Q.21

The following reaction is known as



1. Wolff-Kishner reduction

2. Rosenmund reduction reaction

3. Birch reduction reaction

4. Hell-Volhard-Zelinsky reaction

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which of the following is/are not considered as transition elements?

Ag, Mo, Zn, Hg, W

1. Ag and Zn

2. Mo and Zn

3. Hg and Zn

4. Hg and Mo

Options 1. 1

2. 2

3. 3

4. 4

Q.23 The compound which oxidizes slowly in the presence of air to give extremely poisonous phosgene gas is

1. Dichloromethane
2. Chloroform
3. Carbon tetrachloride
4. Iodomethane

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Mercury cell used in hearing aid devices contains anode comprising of

1. Copper-mercury amalgam
2. Zinc-mercury amalgam
3. Platinum-mercury amalgam
4. Carbon-mercury amalgam

Options 1. 1

2. 2
3. 3
4. 4

Q.25 What will be the order of a reaction if its rate is expressed as

$$\text{Rate} = k[A]^{1/2}[B]^{3/2}$$

1. 2
2. 1
3. 0
4. 3

Options 1. 1

2. 2
3. 3
4. 4

Q.26

The structure of the gluconic acid obtained from oxidation of glucose is

1. $\begin{array}{c} \text{COOH} \\ | \\ (\text{CHOH})_4 \\ | \\ \text{CH}_2\text{OH} \end{array}$
2. $\begin{array}{c} \text{COOH} \\ | \\ (\text{CHOH})_4 \\ | \\ \text{CH}_2\text{COOH} \end{array}$
3. $\begin{array}{c} \text{CHO} \\ | \\ (\text{CHOH})_4 \\ | \\ \text{COOH} \end{array}$
4. $\begin{array}{c} \text{COOH} \\ | \\ (\text{CHOH})_4 \\ | \\ \text{COOH} \end{array}$

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The increasing order of osmotic pressure of 0.1 M aqueous solution of the following is:

- (A) MgCl_2
- (B) NaCl
- (C) NH_2CONH_2
- (D) $\text{K}_3[\text{Fe}(\text{CN})_6]$

Choose the correct answer from the options given below:

1. (A), (B), (D), (C)
2. (A), (B), (C), (D)
3. (C), (B), (A), (D)
4. (D), (A), (B), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 The standard potential of the cell ($E_{\text{cell}}^{\ominus}$) is related to the equilibrium constant (K_C) as

1. $E_{\text{cell}}^{\ominus} = \frac{2.303 nRT}{F} \log K_C$
2. $E_{\text{cell}}^{\ominus} = \frac{2.303 nF}{RT} \log K_C$
3. $E_{\text{cell}}^{\ominus} = \frac{2.303 RT}{nF} \times K_C$
4. $E_{\text{cell}}^{\ominus} = \frac{2.303 RT}{nF} \log K_C$

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Which of the following statements is not correct with reference to a catalyst?

1. It decreases the activation energy of a reaction.
2. It provides an alternative reaction pathway.
3. It catalyzes the forward and backward reaction to the same extent.
4. It alters the Gibbs free energy for a reaction.

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Arrange the following compounds in increasing order towards Nucleophilic addition reactions:

- (A) Propanal
- (B) Benzaldehyde
- (C) Acetone
- (D) Di-tert-butyl ketone

Choose the correct answer from the options given below:

1. (D), (C), (B), (A)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 In alcohols, carbon and oxygen atom are

1. sp^3 and sp^3 hybridised respectively
2. sp^2 and sp^2 hybridised respectively
3. sp^3 and sp^2 hybridised respectively
4. sp^2 and sp^3 hybridised respectively

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which of the following compounds will be colored in the solid state?

1. Ag_2SO_4
2. CuF_2
3. ZnF_2
4. Cu_2Cl_2

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which of the following statements are correct?

- (A) Solutions of acids, bases and salts are the examples of electrolytic conductors.
- (B) Solutions of cane sugar, glycerin, alcohol are examples of non-electrolytes.
- (C) Oxidation takes place at cathode while reduction takes place at anode in an electrolytic cell.
- (D) Silicon and germanium are used in semiconductor devices.

Choose the correct answers from the options given below

- 1. (A), (C) and (D) only
- 2. (B), (C) and (D) only
- 3. (A), (B) and (D) only
- 4. (A), (B) and (C) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.34 A cell is written as



Calculate its E_{cell} if $E_{\text{cell}}^{\ominus} = 1.25 \text{ V}$ at 298 K

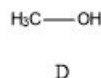
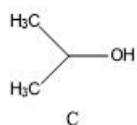
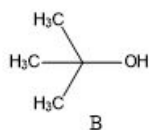
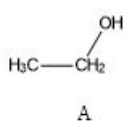
- 1. 1.28 V
- 2. 1.25 V
- 3. 1.19 V
- 4. 1.30 V

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.35

Arrange the following alcohols in decreasing order of acidity :



Choose the **correct** answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (B), (A), (C), (D)
- 3. (A), (B), (D), (C)
- 4. (D), (A), (C), (B)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.36 Which of the following is a fat soluble vitamin?

- 1. Thiamine
- 2. Riboflavin
- 3. Ascorbic acid
- 4. Vitamin K

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.37 Match List-I (reagent) with List-II (Transformations)

List-I (Reducing Agent) List-II (Transformation)

- (A) Sodium Borohydride (I) Aldehyde to alkane
- (B) SnCl_2 and HCl (II) aldehydes to Primary alcohols

List-I (Reducing Agent) List-II (Transformation)

(C) Zn-Hg and conc. HCl (III) Nitriles to Imine

(D) DIBAL-H (IV) Ester to aldehyde

Choose the correct answer:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Which of the following statements is correct for enantiomers?

- (A) They exist as a pair of non-superimposable mirror images.
(B) They possess identical physical properties.
(C) They differ with respect to the rotation of plane polarized light.
(D) A mixture containing a 1:1 ratio of enantiomers is optically inactive.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

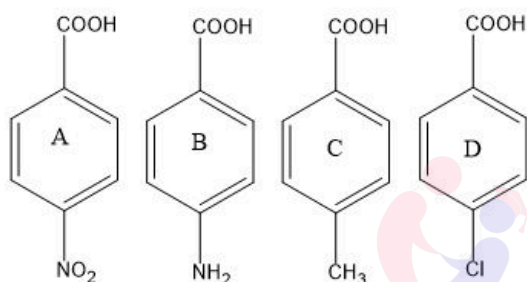
2. 2

3. 3

4. 4

Q.39

Arrange the following compounds in increasing order of their acidic strength ?



Choose the **correct** answer from the options given below:

1. (C), (B), (D), (A)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (A), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Which of the following statements are correct?

- (A) In ideal solutions, the enthalpy of mixing of the pure components to form the solution is zero.
(B) The volume of the solution is equal to the sum of the volumes of the two components.
(C) In binary ideal solutions, attractive forces between A-A and B-B are nearly equal to those between A-B.
(D) In negative Raoult's law deviation, attractions between A-B are weaker than those of A-A and B-B attractions.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (A), (B), and (D)
4. (A), (B), and (C) only

Options 1. 1

2. 2

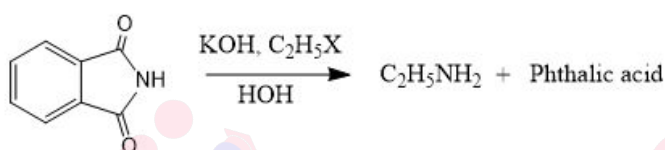
3. 3

4. 4

Q.41 Read the passage carefully and answer the Questions

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecules. Aliphatic amines are a stronger bases due to $+I$ effect of an alkyl groups. Intermolecular hydrogen bonding in primary and secondary amines affects the boiling points of amines. The intermolecular association is more in primary amines than in secondary amines as there are more number of hydrogen atoms available for hydrogen bond formation. Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in an acidic medium. Gabriel synthesis is used for the preparation of primary amines. Primary amines react with nitrous acid to give unstable diazonium-salts which can be used to synthesize a variety of compounds.

The reaction is an example of



1. Hoffmann reaction

2. Gabriel synthesis

3. Kolbe reaction

4. Sandmeyer reaction

Options 1. 1

2. 2

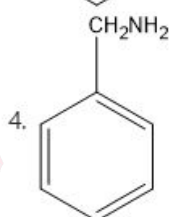
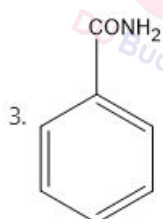
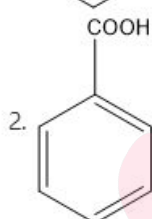
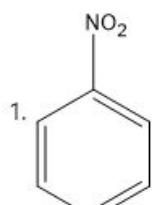
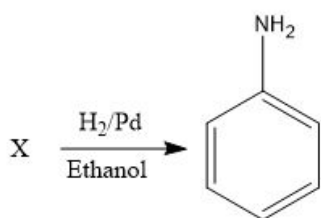
3. 3

4. 4

Q.42 Read the passage carefully and answer the Questions

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecules. Aliphatic amines are a stronger bases due to $+I$ effect of an alkyl groups. Intermolecular hydrogen bonding in primary and secondary amines affects the boiling points of amines. The intermolecular association is more in primary amines than in secondary amines as there are more number of hydrogen atoms available for hydrogen bond formation. Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in an acidic medium. Gabriel synthesis is used for the preparation of primary amines. Primary amines react with nitrous acid to give unstable diazonium-salts which can be used to synthesize a variety of compounds.

X in the following reaction is



Options 1. 1

2. 2

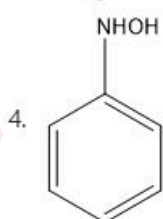
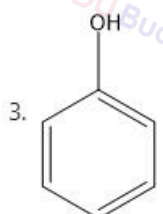
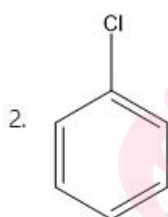
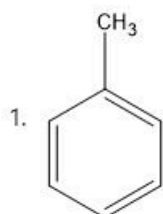
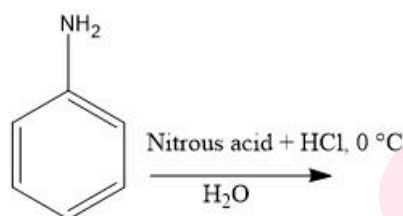
3. 3

4. 4

Q.43 Read the passage carefully and answer the Questions

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecules. Aliphatic amines are stronger bases due to +I effect of an alkyl group. Intermolecular hydrogen bonding in primary and secondary amines affects the boiling points of amines. The intermolecular association is more in primary amines than in secondary amines as there are more number of hydrogen atoms available for hydrogen bond formation. Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in an acidic medium. Gabriel synthesis is used for the preparation of primary amines. Primary amines react with nitrous acid to give unstable diazonium-salts which can be used to synthesize a variety of compounds.

The product in the following reaction is



Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the passage carefully and answer the Questions

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecules. Aliphatic amines are stronger bases due to +I effect of an alkyl groups. Intermolecular hydrogen bonding in primary and secondary amines affects the boiling points of amines. The intermolecular association is more in primary amines than in secondary amines as there are more number of hydrogen atoms available for hydrogen bond formation. Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in an acidic medium. Gabriel synthesis is used for the preparation of primary amines. Primary amines react with nitrous acid to give unstable diazonium-salts which can be used to synthesize a variety of compounds.

The compound having the highest boiling point is

1. n-pentane

2. Diethylamine

3. n-Butanol

4. n-Butylamine

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the passage carefully and answer the Questions

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecules. Aliphatic amines are stronger bases due to +I effect of an alkyl group. Intermolecular hydrogen bonding in primary and secondary amines affects the boiling points of amines. The intermolecular association is more in primary amines than in secondary amines as there are more number of hydrogen atoms available for hydrogen bond formation. Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in an acidic medium. Gabriel synthesis is used for the preparation of primary amines. Primary amines react with nitrous acid to give unstable diazonium-salts which can be used to synthesize a variety of compounds.

Identify the amine possessing least basic strength

1. Aniline
2. p-Nitroaniline
3. p-Toluidine
4. p-Methoxyaniline

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the Questions

According to Valence Bond theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on. These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from many shortcomings. The Crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

Which of the following represents the correct pair?

1. d^2sp^3 - spin free complex
2. inner orbital complex - low spin complex
3. outer orbital complex - low spin complex
4. d^2sp^3 - outer orbital complex

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the Questions

According to Valence Bond theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on. These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from many shortcomings. The Crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

Which of the following is not true about the limitations of VBT

1. It does not give quantitative interpretation of magnetic data.
2. It does not explain hybridisation to yield a set of equivalent orbitals
3. It does not explain the colour exhibited by coordination compounds.
4. It does not give a quantitative interpretation of the thermodynamic or kinetic stabilities of coordination compounds.

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the Questions

According to Valence Bond theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd

orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on. These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from the many shortcomings. The Crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM. Predict the geometry of the complex ion?

1. tetrahedral
2. square planar
3. trigonal pyramidal
4. distorted octahedral

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the Questions

According to Valence Bond theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on. These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from the many shortcomings. The Crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

Select the incorrect statement about the Crystal field splitting in octahedral complexes.

1. $d_{x^2-y^2}$ and d_{z^2} orbitals point towards the axes along the direction of the ligand will experience less repulsion
2. d_{xy} , d_{yz} and d_{zx} orbitals are directed between the axes will be lowered in energy
3. the degeneracy of the d orbitals has been removed to yield three orbitals of lower energy, t_{2g} set
4. degeneracy of the d orbitals has been removed to yield two orbitals of higher energy, e_g set

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the Questions

According to Valence Bond theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on. These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from the many shortcomings. The Crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

Which among the following are diamagnetic complexes?

$[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{CoF}_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$

1. $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{CoF}_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$
2. $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Fe}(\text{CN})_6]^{4-}$
3. $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{CoF}_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$
4. $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$

Options 1. 1

2. 2
3. 3
4. 4